Identifying IPC Sections from Indian Case Facts, and the Move to the Bharatiya Nyaya Sanhita

A comparison of lexical and transformer models on the ILSI dataset, with an IPC → BNS transition analysis

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*Unless stated otherwise, every number is measured on the ILSI **test** set; every hyperparameter, threshold strategy and threshold was chosen on the **dev** set only.*

1. Abstract

We predict which Indian Penal Code (IPC) sections apply to an Indian criminal judgment, a multi-label classification task over 100 sections, using the ILSI dataset (about 66,000 judgments). We compare a statute-keyword baseline, TF-IDF with logistic regression, and two fine-tuned BERT models (general-domain bert-base-uncased and the legal-domain InLegalBERT).

TF-IDF with logistic regression is the best model (micro-F1 0.691, macro-F1 0.585). It beats the best BERT model, InLegalBERT (0.520 / 0.445), by 17 micro-F1 points (paired bootstrap, $p < 0.001$). A diagnostic that only extracts "section N" citations from the text scores 0.701 micro-F1 without learning anything: much of ILSI's label signal is section numbers the judgments already cite, which whole-document lexical models exploit and 512-token BERT only partly sees. On the 19% of test judgments that cite no gold section, all models are weak (0.31–0.35) but clearly beat the citation diagnostic (0.07), so they do infer offences from facts.

We also map the 100 labels to the Bharatiya Nyaya Sanhita (BNS), which replaced the IPC on 1 July 2024: 84 map one-to-one and four unverifiable mappings are withheld. Renumbering, new offences and citation-dependent models mean that relabelling alone will not keep such a system valid. A themed web demo shows predictions together with their BNS counterparts.

2. Introduction and problem statement

Legal statute identification asks: given the facts or text of a case, which provisions of law apply? For Indian criminal law this has meant sections of the IPC, for example §302 (murder) with §34 (common intention). Automating it could help legal research, case triage and legal aid. It is also a hard multi-label problem:

- documents carry about four sections each;
- sections co-occur strongly (§147/§148/§149 for rioting; §323/§506 for hurt and intimidation);
- frequency is very skewed;
- neighbouring sections differ in fine legal details (weapon used, severity of injury, intention).

Task. Given the text of a judgment, output the set of applicable IPC sections from a fixed list of 100.

Questions.

1. How do lexical baselines compare with fine-tuned BERT models, and does legal-domain pre-training help?
2. What do the models actually rely on?
3. What would it take to keep such a model useful after the switch from the IPC to the BNS?

Metrics.

- **micro-F1**, dominated by frequent sections;
- **macro-F1**, every section weighted equally;
- micro precision and recall, samples-F1, and mean per-label F1 in head, mid and tail frequency buckets.

3. Dataset and exploratory analysis

ILSI (Indian Legal Statute Identification; Paul, Goyal & Ghosh, AAAI 2022; Zenodo record 6053791) contains Indian court judgments, each labelled with the IPC sections it is linked to, plus the text of the 100 sections. We used the official splits and verified the MD5 checksums of the downloaded files.

Table 1. Dataset statistics.

	Train	Dev	Test
Documents	42,835	10,200	13,039
Labels per document (mean)	3.70	3.91	3.90
Words per document (median / 95th percentile)	458 / 4,128	579 / 4,486	567 / 4,271
Documents longer than ~400 words (≈ BERT's 512 tokens)	54%	60%	59%

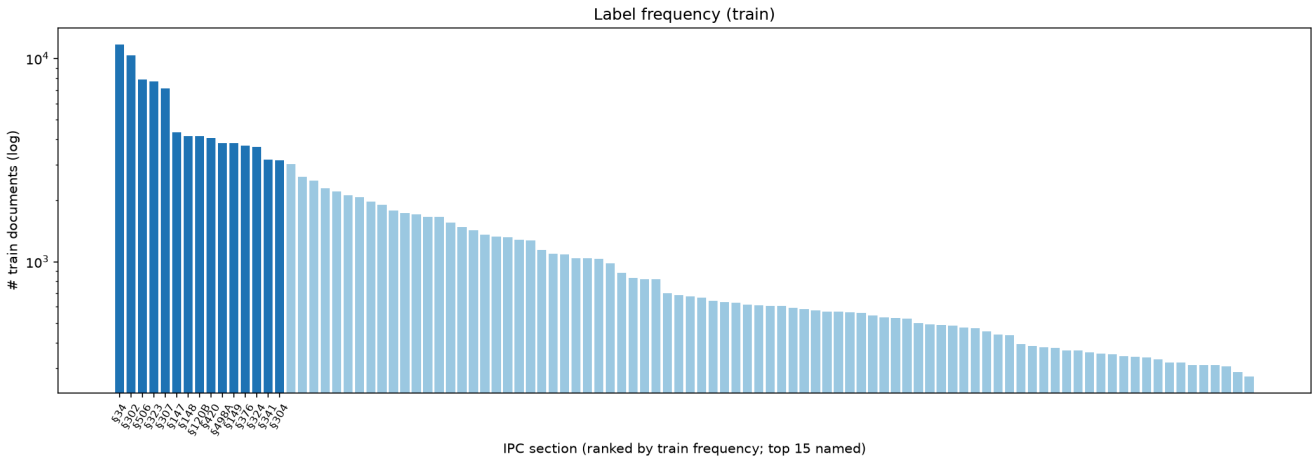


Figure 1. *Label frequency in training data (log scale); the 15 most frequent sections are named.*

Imbalance. §34 occurs in 11,751 training documents, §450 in only 272 (43×). We split labels into tertiles by training frequency:

- **head:** 34 labels with 1,329–11,751 documents each, carrying 75% of all label occurrences;
- **mid:** 33 labels, 544–1,322 documents;
- **tail:** 33 labels, 272–533 documents, 8% of occurrences.

These buckets are used throughout the analysis.

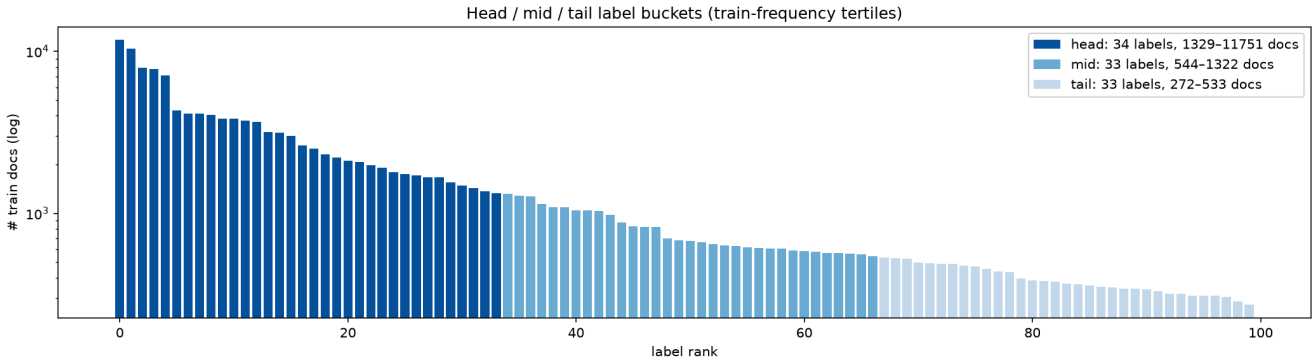


Figure 2. *Head / mid / tail label buckets.*

Length. More than half of all documents exceed the ~400 words that fit into BERT's 512-token input, so truncation matters (Figure 3).

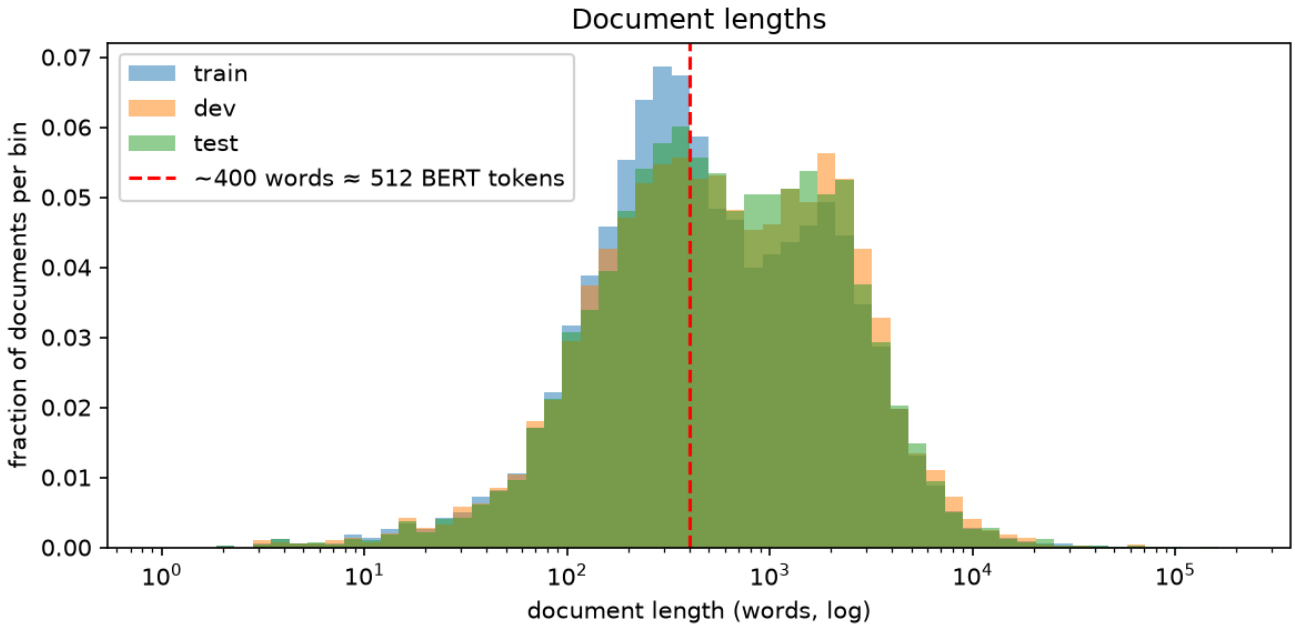


Figure 3. *Document length distribution; the red line marks ~400 words ≈ 512 BERT tokens.*

Co-occurrence. The strongest pairs are §147 & §148 (Jaccard 0.71), §148 & §149, and §302 & §34 (3,650 training documents).

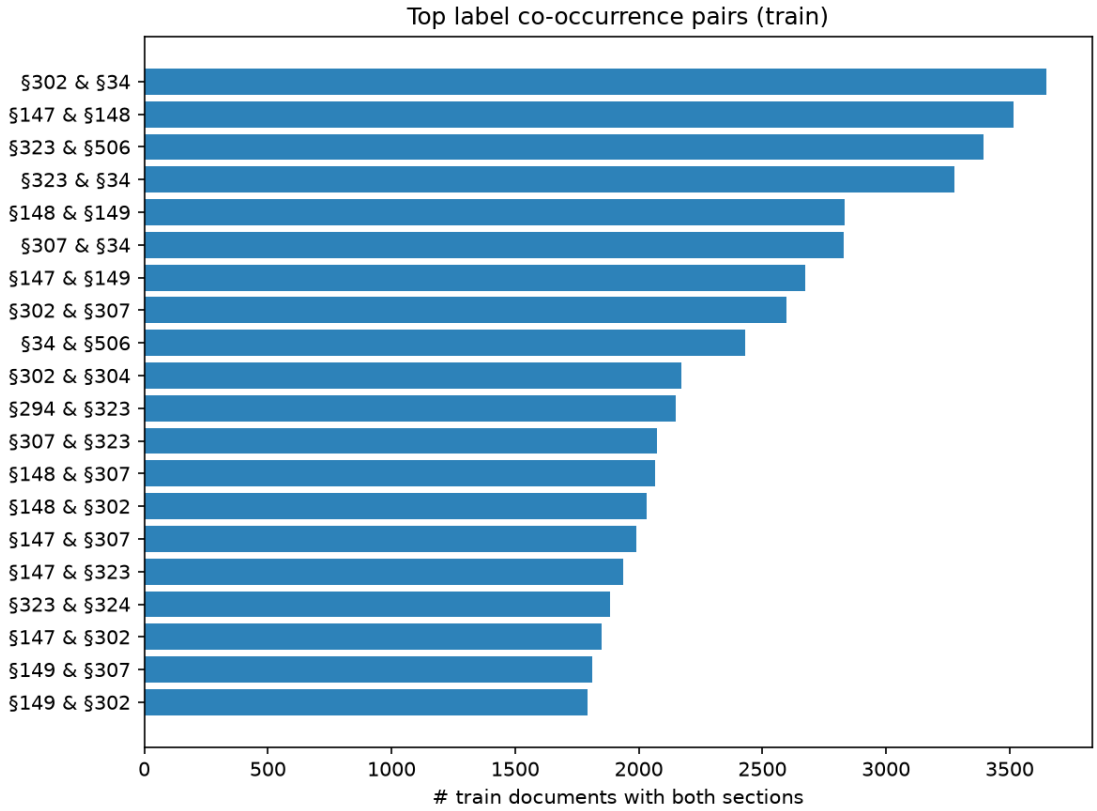


Figure 4. *The 20 most frequent co-occurring label pairs.*

Data-quality findings. These matter for how the results should be read:

1. **Citations are not masked.** A list-aware citation parser finds a gold section cited verbatim ("Section 302 I.P.C.", "u/s 147, 148, 149") in **81%** of test documents: within the first ~400 words in 65%, only later in 16%, never in 19%.
2. **Non-offence and repealed sections are labels.** §2–§5 describe where the Code applies; §13 was repealed in 1950 and §161/§164 in 1988.
3. **Likely mis-linked citations.** In at least one test judgment, the gold labels §156, §173 and §200 correspond to *Code of Criminal Procedure* sections cited in the text ("Section 156(3) Cr.P.C."). Some labels may therefore come from other Acts' citations. This was observed in examples, not measured at scale.
4. **Near-empty documents.** 0.6% of documents have fewer than 10 words (for example only the judge's name).

4. Methods

Every model writes the same outputs (dev and test probabilities in one fixed label order, tuned thresholds, metrics), and all of them are scored by a single unit-tested evaluation module, so the results are directly comparable.

4.1 Baselines (laptop, Apple M1, 8 GB)

- **Keyword.** TF-IDF over the 100 statute texts (stop words and numbers removed). Each section keeps its top-k distinctive terms (k = 5, chosen on dev from {5, 10, 15, 25}). A document's score for a section is the weighted share of those terms it contains. No training documents are used.
- **TF-IDF + logistic regression.** Word 1–2-gram TF-IDF (100,000 features, sublinear term frequency, min_df 3, max_df 0.9) with one-vs-rest logistic regression (liblinear, class-balanced weights). The regularisation strength was chosen on dev: C = 1 (dev micro-F1 0.685) over C = 0.5 (0.661). The planned C ∈ {4, 10} and a linear-SVM variant were dropped because each C value needed about 40 minutes on the laptop.
- **Citation diagnostic.** Predicts §N whenever the text cites "section N" or "N IPC". It is not a model; it measures how much of the task can be solved by reading citations.

4.2 BERT models (Google Colab, NVIDIA A100)

- **Encoders.** `bert-base-uncased` and `law-ai/InLegalBERT` (LegalBERT further pre-trained on Indian court judgments; Paul et al., 2023). Each gets a 100-way sigmoid output layer.
- **Training.** 4 epochs, learning rate $2 \cdot 10^{-5}$ with 10% warm-up and linear decay, batch 16 × gradient accumulation 2, weight decay 0.01, mixed precision, seed 42. The epoch with the best dev micro-F1 (at threshold 0.5) is kept. Each run took about 13 minutes.
- **Truncation.** *head* keeps the first 510 tokens. *head+tail* keeps the first 128 and the last 382, because judgments often end with the operative order.
- **Class imbalance.** Binary cross-entropy with a per-label positive weight = negatives / positives, clipped to [1, 30] (67 of the 100 labels hit the cap).
- **Validation before GPU use.** The training notebook was run end-to-end on the laptop with a tiny BERT model first. This found and fixed a crash in head+tail tokenisation.

4.3 Thresholding

Three strategies are evaluated, all tuned on dev:

- a fixed threshold of 0.5;
- one global threshold (0.05–0.95, step 0.01) maximising micro-F1;
- per-label thresholds maximising each label's F1, falling back to the global threshold for labels with fewer than 5 dev positives.

A document with no predicted label receives its single most likely label. Each model is reported at the strategy with the highest *dev* micro-F1, which was per-label for every learned model.

5. Results

Table 2. Main results (test set; each model at its dev-selected threshold strategy; the citation diagnostic is shown for reference).

Model	Thresholds	Micro-F1	Macro-F1	Micro-P	Micro-R	Samples-F1	Head F1	Mid F1	Tail F1
Keyword (statute terms)	global	0.176	0.107	0.162	0.191	0.131	0.176	0.083	0.061
Citation diagnostic	—	*0.701*	*0.627*	*0.703*	*0.699*	*0.640*	*0.759*	*0.621*	*0.497*
TF-IDF + LR	per-label	**0.691**	**0.585**	0.707	0.675	**0.654**	0.708	0.582	0.463
BERT-base, head	per-label	0.497	0.417	0.429	0.590	0.482	0.546	0.409	0.293
InLegalBERT, head	per-label	0.520	0.445	0.466	0.589	0.500	0.557	0.438	0.336
InLegalBERT, head+tail	per-label	0.511	0.436	0.458	0.578	0.495	0.550	0.424	0.329

All threshold strategies for every model are in `tables/results_all_strategies.csv`.

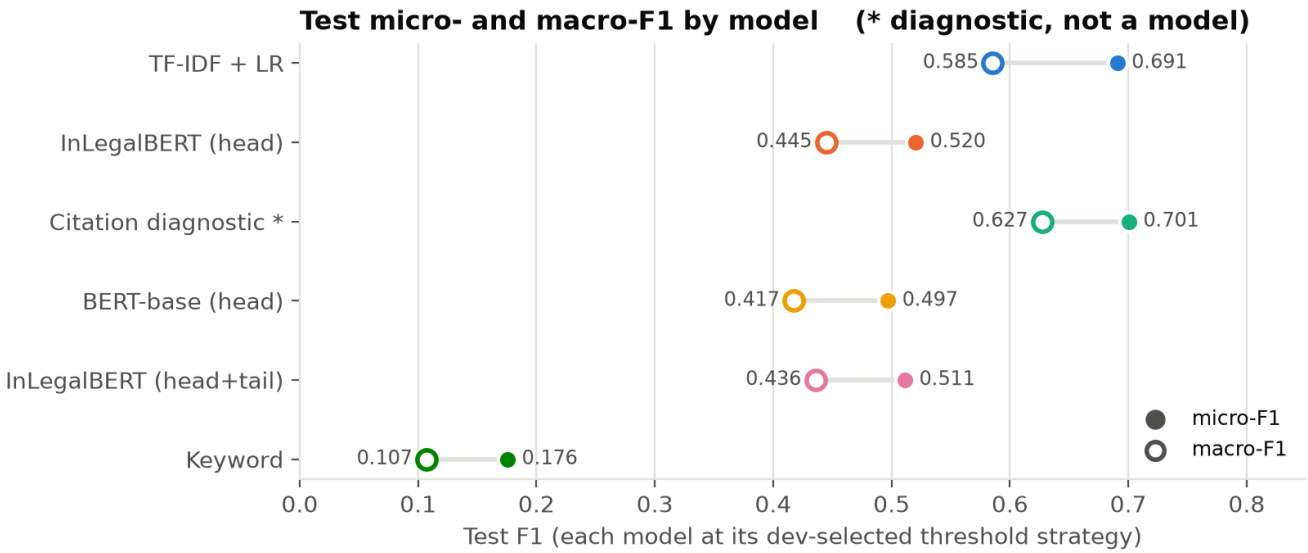


Figure 5. *Test micro-F1 (filled) and macro-F1 (hollow) for every model.*

Thresholds matter, especially for BERT (Figure 6). The class weighting pushes BERT's probabilities up, so at 0.5 InLegalBERT reaches only 0.394 micro-F1 (precision 0.26). A tuned global threshold (0.77) gives 0.504 and per-label thresholds give 0.520. For TF-IDF the steps are 0.647 → 0.682 → 0.691.

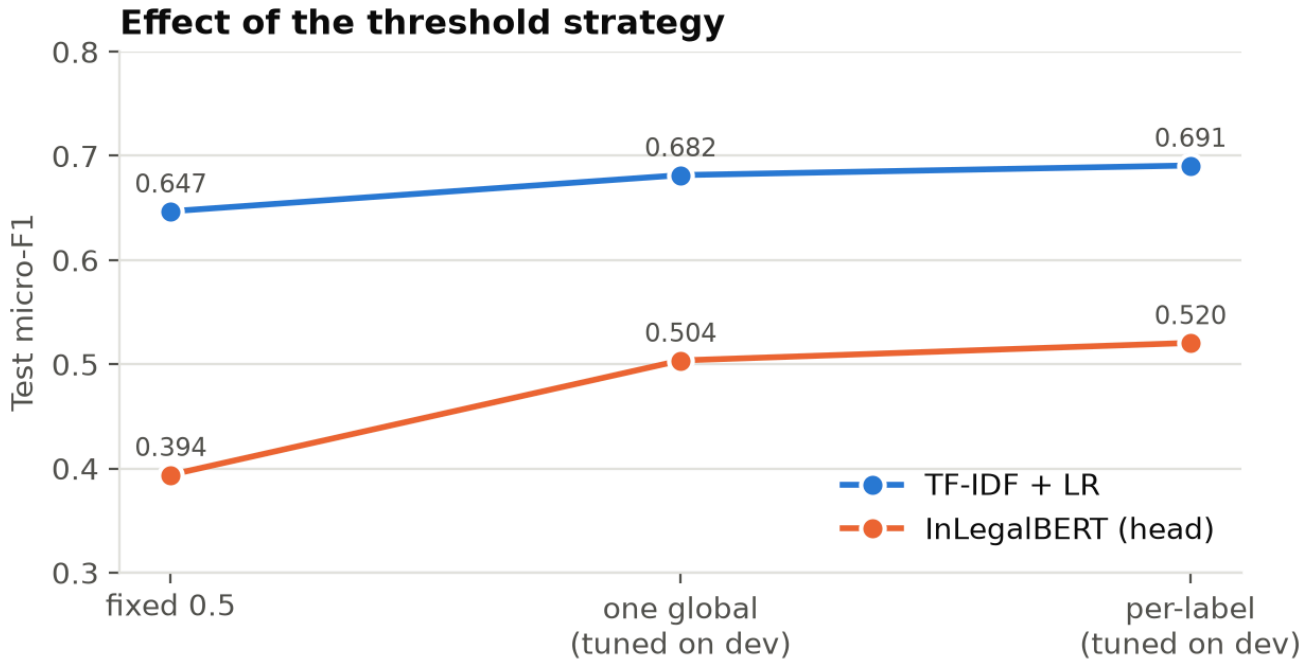


Figure 6. *Effect of the threshold strategy on test micro-F1.*

Legal pre-training helps: InLegalBERT beats bert-base-uncased by +2.4 micro-F1 and +2.7 macro-F1, with the largest gain on tail labels (+0.04). **Head+tail truncation did not help** (−0.9 micro-F1). Training was stable, and two of the three runs were still improving slightly at epoch 4 (Figure 7).

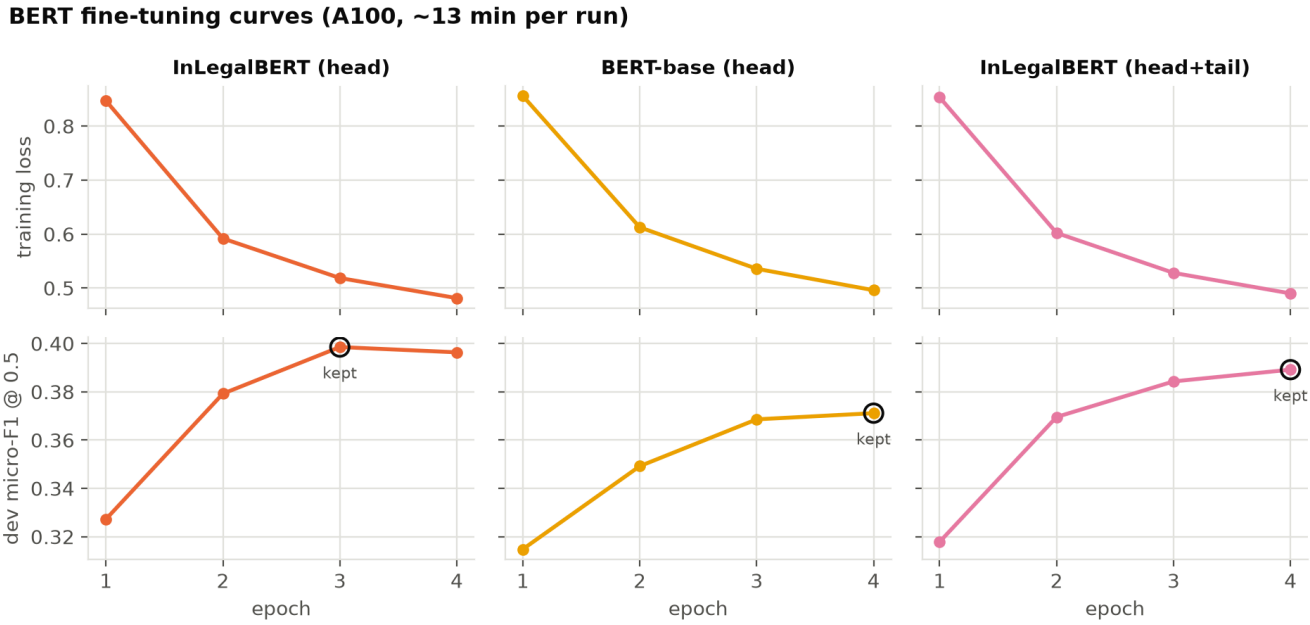


Figure 7. *Training loss (top) and dev micro-F1 at threshold 0.5 (bottom) per epoch; the circled epoch was kept.*

6. Analysis

6.1 Label frequency

The ranking of models is identical in every frequency bucket, and the learned models lose 0.22–0.26 F1 from head to tail labels (Figure 8, Table 3). BERT's gap to TF-IDF is roughly constant (−0.15 head, −0.14 mid, −0.13 tail), so BERT is not specifically better on rare labels. Per-label F1 correlates with log training frequency (Pearson r = 0.63 for TF-IDF, 0.52 for InLegalBERT, Figure 9). For both models the hardest labels are the non-offence sections §2, §3 and §4.

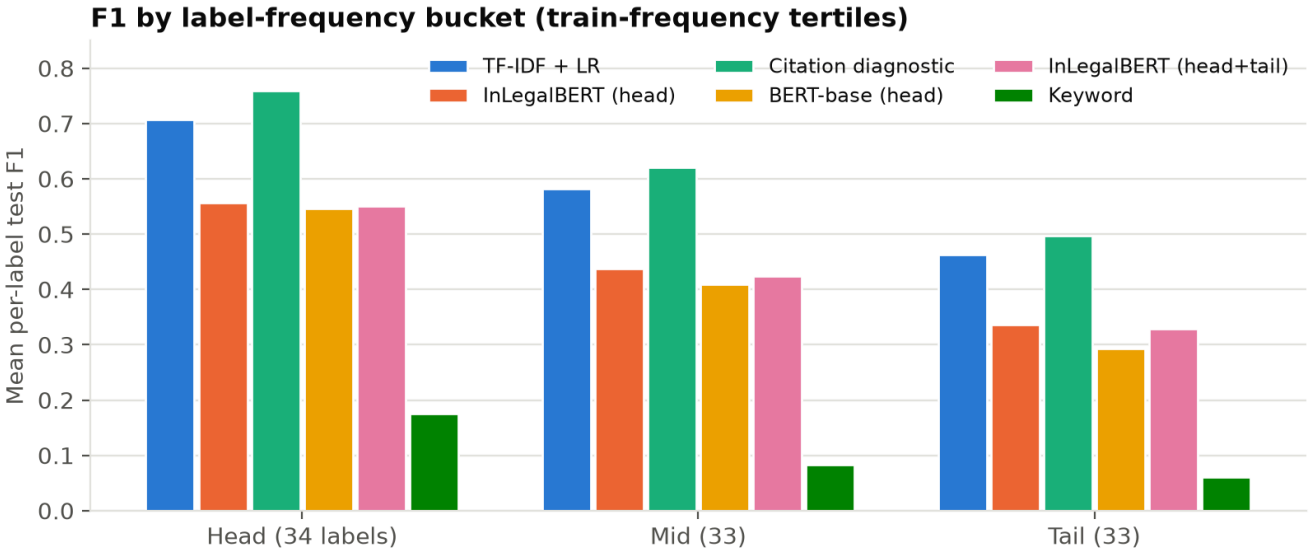


Figure 8. *Mean per-label test F1 by frequency bucket.*

Table 3. Mean per-label F1 by bucket.

Model	Head F1	Mid F1	Tail F1
TF-IDF + LR	0.708	0.582	0.463
InLegalBERT (head)	0.557	0.438	0.336
Citation diagnostic	0.759	0.621	0.497
BERT-base (head)	0.546	0.409	0.293
InLegalBERT (head+tail)	0.550	0.424	0.329
Keyword	0.176	0.083	0.061

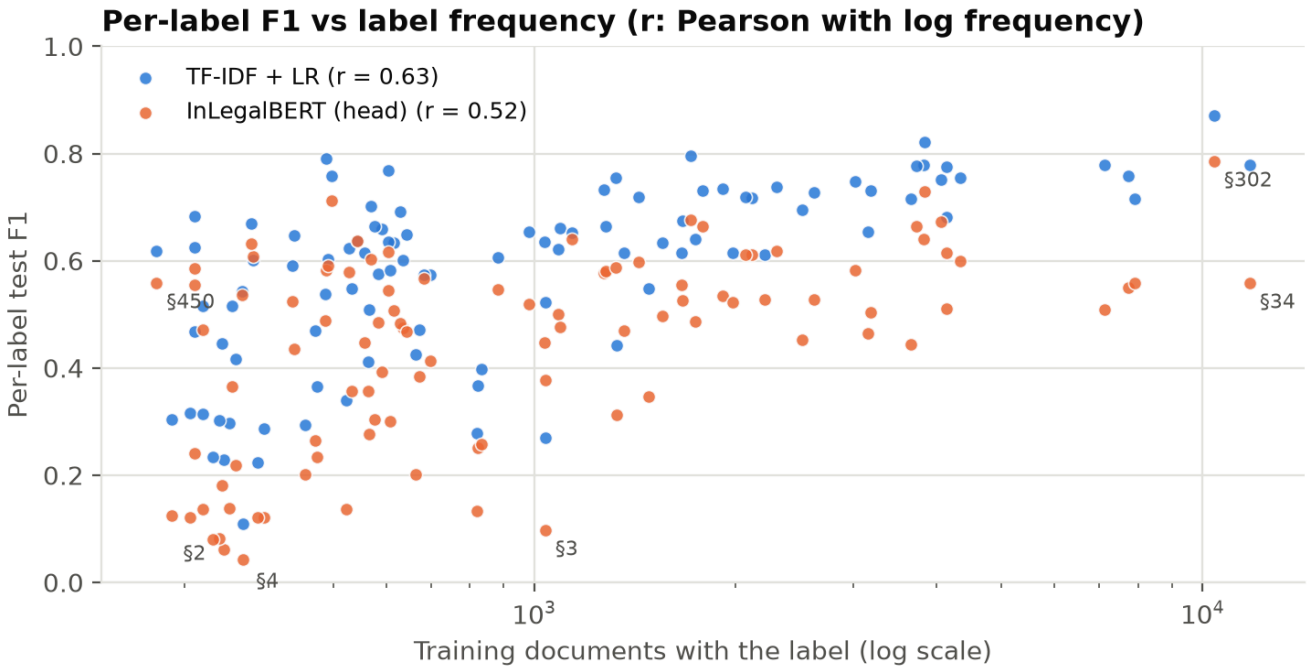


Figure 9. *Per-label test F1 against the number of training documents (log scale).*

InLegalBERT matches or beats TF-IDF on only **1 of 100 labels** (§337, +0.007). Its largest losses (−0.21 to −0.36 F1) are §437, §465, §324, §307, §326 and §34: sections where the cited number is strong evidence and neighbouring sections read alike in prose. Full per-label scores are in `tables/per_label_f1.csv`.

6.2 Confusions between related sections

Errors concentrate in legally adjacent, often co-charged families, and InLegalBERT makes them 3–4× more often than TF-IDF (Figure 10):

- §34 is falsely added when §302, §307 or §149 is the true section (41% of §149 documents for BERT, 13% for TF-IDF);
- BERT mixes up the murder ladder §302/§304/§307;
- BERT mixes up the hurt ladder §323/§324/§325 (§324 is predicted in 15% of §323 documents, against 2% for TF-IDF).

Rioting (§147/§148/§149) and cheating vs criminal breach of trust (§420/§406) are confused less often (BERT 8–17%).

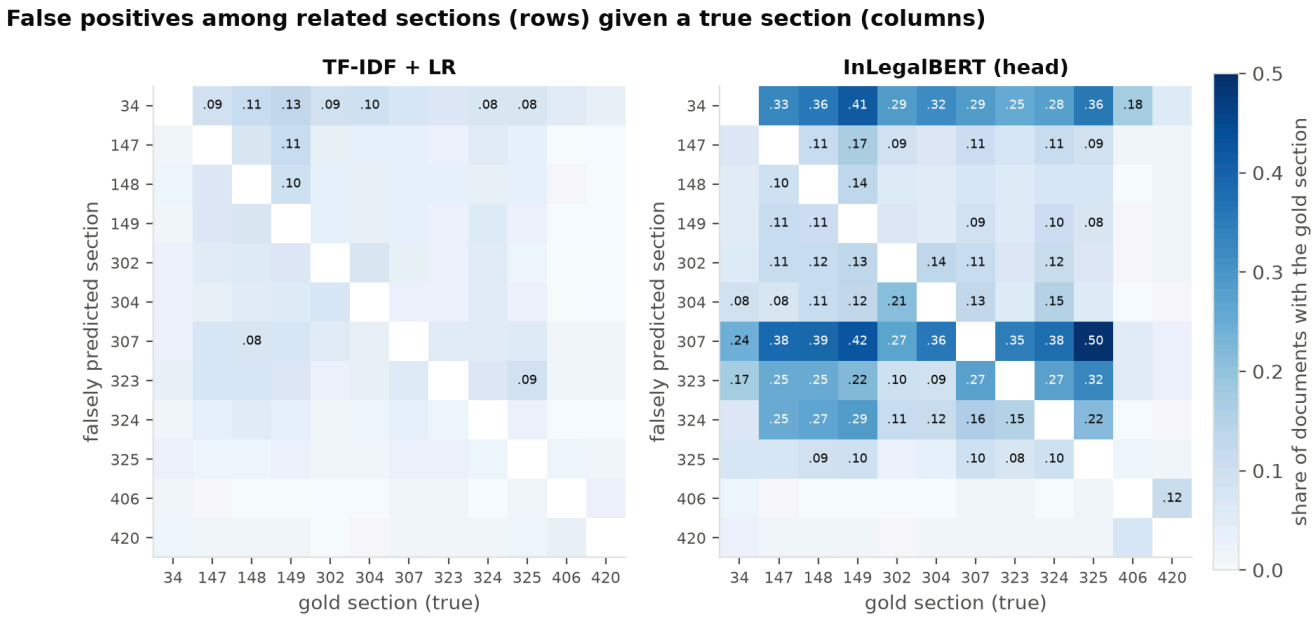


Figure 10. *Share of documents with a given true section (column) in which a related section (row) is falsely predicted.*

6.3 Why BERT trails: where the citations are

Splitting the test set by where a gold section is first cited explains most of the gap (Figure 11, Table 4):

1. **Truncation.** Late citations are invisible to BERT, and its micro-F1 drops from 0.59 to 0.43, while TF-IDF reads the whole document (0.76 → 0.69).
2. **Citation copying.** Even when the citation is inside its window, BERT does not reliably "copy" it (0.59 vs 0.78 for the regex).
3. **Facts only.** Without any citation, every model is weak, but both learned models clearly beat the regex (0.35 and 0.31 vs 0.07). This subset is closest to the intended task of inferring statutes from facts.

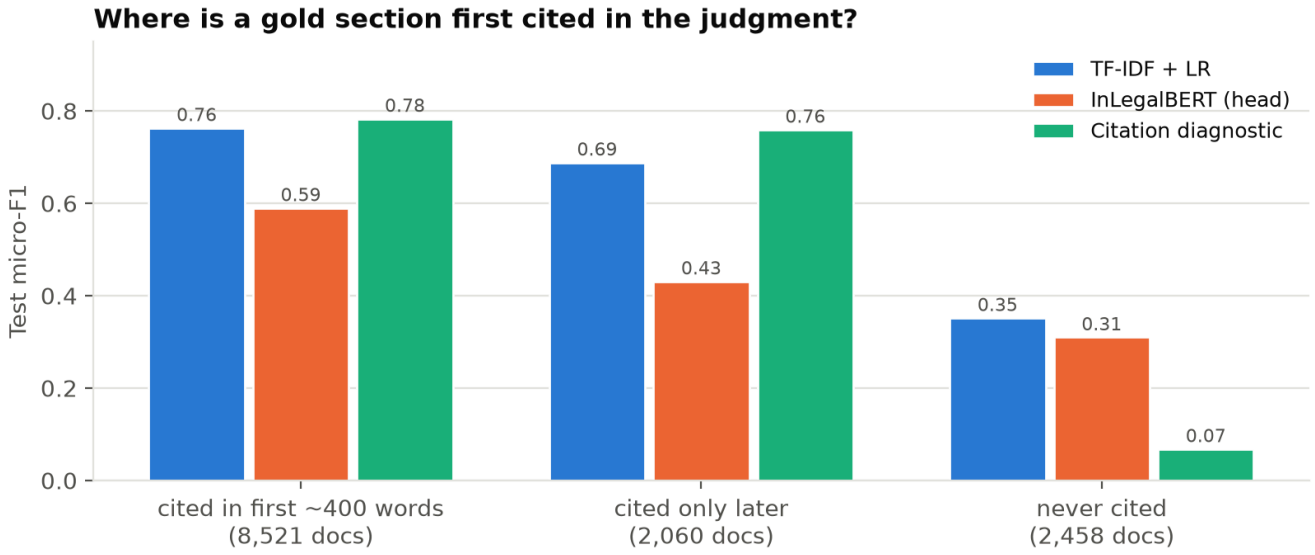


Figure 11. *Test micro-F1 split by where a gold section is first cited in the judgment.*

Table 4. Micro-F1 by citation position.

Documents where a gold section is...	Docs	TF-IDF + LR	InLegalBERT (head)	Citation diagnostic
named in first ~400 words	8521	0.763	0.590	0.783
named only after ~400 words	2060	0.688	0.431	0.759
never named	2458	0.352	0.310	0.068

6.4 Error cases

370 test documents have document-level F1 ≥ 0.75 for TF-IDF and ≤ 0.25 for BERT; only 32 show the reverse. Eight of each were sampled at random (seed 42) and read in full; per-case comments are in the analysis notebook (`notebooks/02_analysis.ipynb`).

- **BERT fails, TF-IDF succeeds.** Five of eight are long judgments whose relevant citations sit beyond the 510-token window, or whose opening is procedural (a service-law writ, a correction application). In the rest, BERT predicts a semantically related cluster instead of the cited sections, e.g. §229A (bail-jumping) for a bail order that cites §§363/366/376/506 in its first 40 words.
- **TF-IDF fails, BERT succeeds.** All eight are short (131–432 words) and seven contain no citation. BERT infers the offence from the narrative: §302 from "sentenced to suffer imprisonment for life", the rioting trio from "five appellants", and bail-jumping from a bail-application context.

6.5 Statistical significance

Table 5. Paired bootstrap over test documents (1,000 resamples, seed 42; 95% confidence intervals).

Metric	TF-IDF + LR [95% CI]	InLegalBERT [95% CI]	Difference [95% CI]	p (two-sided)
micro-f1	0.691 [0.6857, 0.696]	0.520 [0.5155, 0.5249]	-0.170 [-0.1747, -0.1659]	< 0.001
macro-f1	0.585 [0.5779, 0.5917]	0.445 [0.4379, 0.4502]	-0.141 [-0.1469, -0.134]	< 0.001

The difference is large and significant: no resample favoured BERT. This captures sampling variation in the test set, not training variance, because each BERT configuration was trained once.

7. IPC → BNS transition

The full discussion is in `bns_discussion.md`; the mapping is in `tables/ipc_to_bns_mapping.csv`.

Building the mapping. An official MHA/BPR&D correspondence table could not be accessed. Each of the 100 labels was checked row by row against a correspondence table compiled by the Director of CAPT Bhopal (a secondary source, cited in every row).

- 96 rows are confirmed by the source (or, for the three repealed sections, by ILSI's own statute text).
- For the remaining 4 (IPC 143, 427, 294(b), 376(2)) no BNS section could be verified, so their mapping is **withheld** rather than guessed.

Coverage (Figure 12, Table 6):

- 84 labels map one-to-one, carrying 86.8% of test label occurrences.
- 8 changed in scope (e.g. grievous hurt's 20-day threshold became 15 days); §506 splits into BNS 351(2) and 351(3); 3 were repealed long before the BNS.

- Because documents carry about four labels, only **63.4%** of test documents map cleanly in full.
- One-to-one relabelling leaves F1 unchanged by construction (TF-IDF 0.701, InLegalBERT 0.525 on the clean labels). It renames outputs and fixes nothing else.

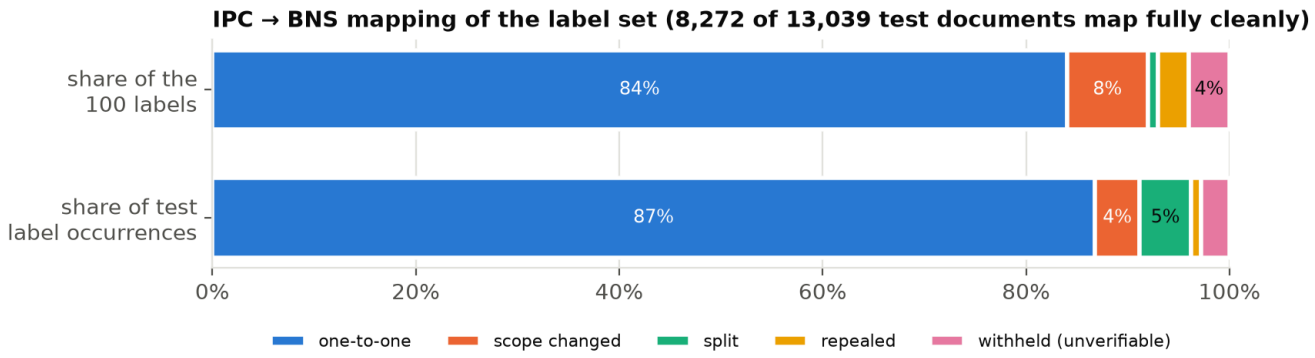


Figure 12. *IPC → BNS mapping types, as a share of labels and of test label occurrences.*

Table 6. Mapping coverage.

Mapping type	Labels	Share of test label occurrences
one-to-one	84	0.868
scope changed	8	0.044
split	1	0.050
repealed	3	0.010
withheld (unverifiable)	4	0.028

What breaks.

- **New offences have no training data:** organised crime (BNS 111), terrorist act (113), snatching (304), mob lynching (103(2)), false promise of marriage (69) and others (Appendix C).
- **The models read citations, and the numbers now mean different things.** For 95 of 100 labels, the TF-IDF model's five strongest features include the section's own number. Under the BNS: BNS 302 punishes hurting religious feelings, BNS 304 is snatching, BNS 307 is a theft offence. A model trained on IPC judgments will misfire on BNS-era text.
- **Both codes run in parallel.** Offences committed before 1 July 2024 continue under the IPC, so a deployed system must handle both codes for years. Meanwhile BNS-era reported judgments will remain scarce.

Update strategies.

1. **Deterministic relabelling,** applied on inputs with section numbers masked.
2. **Fine-tuning on BNS judgments** as they accumulate, with a dual IPC/BNS label space.
3. **Retrieval against statute text,** which needs only the statute's text for a new offence. As a calibration, the fine-tuned InLegalBERT encoder matching case facts to the 100 IPC statute texts reaches recall@5 = 0.29, against 0.05 for random ranking and 0.63 for the supervised classifier (Figure 13). The same experiment against BNS texts is implemented but was not run, because no reliable machine-readable BNS text was available.

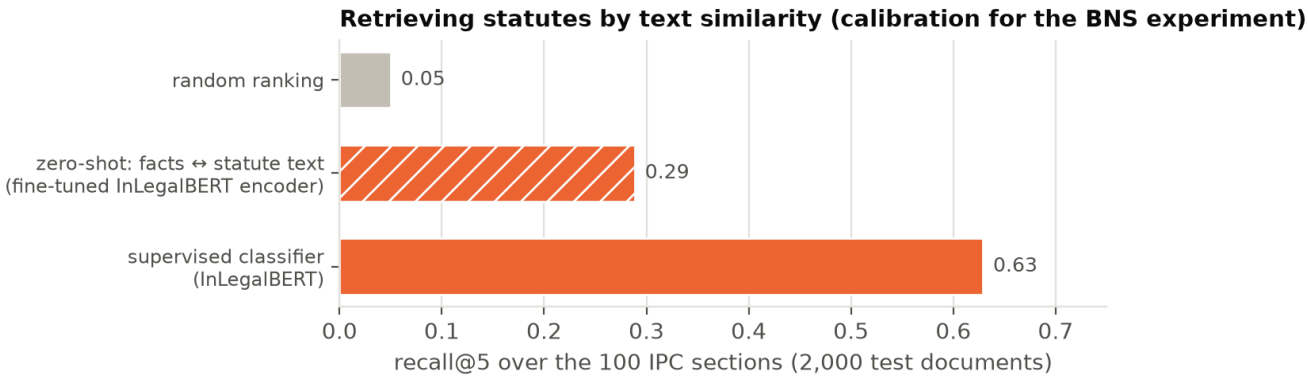


Figure 13. *Retrieving statutes by text similarity (recall@5, 2,000 test documents).*

8. Demo application

A local web application, *Statute Finder*, demonstrates the models (FastAPI back end; hand-built HTML/CSS/JS front end with a law-library theme in dark and light variants).

- **Input:** paste case facts, or load one of three "exhibits" drawn from the test set.
- **Model choice:** fine-tuned InLegalBERT (default) or TF-IDF + LR, each shown with its test scores.
- **Output:** the ten most likely sections, each with its score bar and its own dev-tuned threshold, an "Applies" stamp when the threshold is cleared, and its BNS counterpart marked *verified*, *split*, *scope changed*, *withheld* or *repealed*.
- **Warnings:** sections already cited in the input are flagged, because the models lean on them. A "not legal advice" disclaimer is shown at the top and bottom of the page.

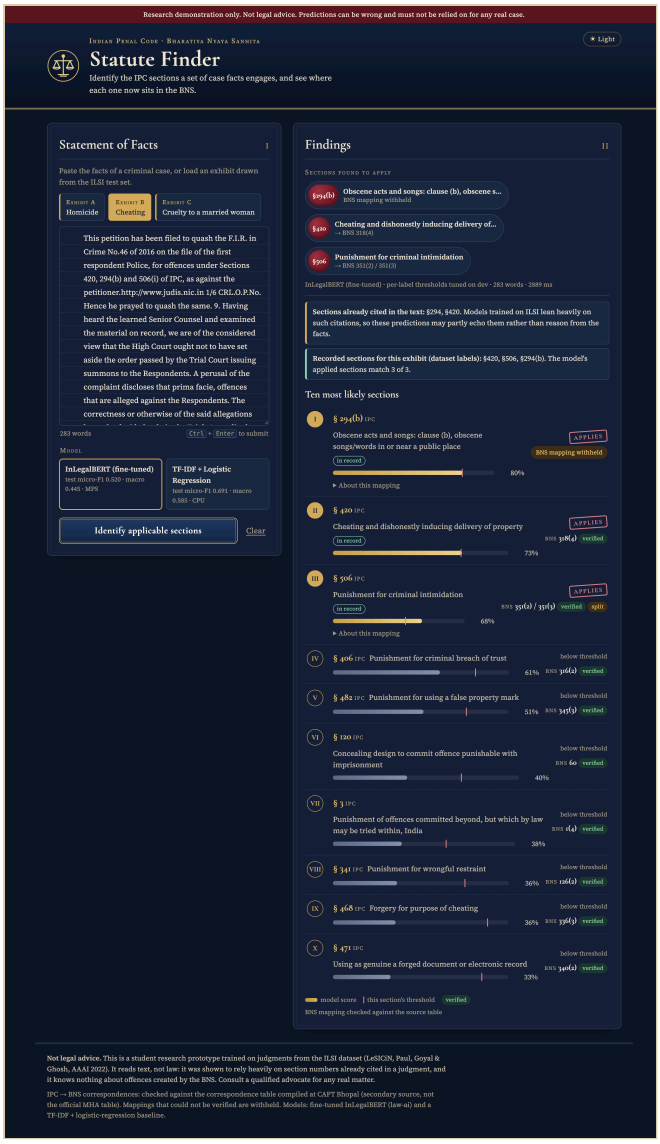


Figure 14. *The demo application (dark theme) on a test-set exhibit: InLegalBERT recovers all three recorded sections and shows their BNS status.*

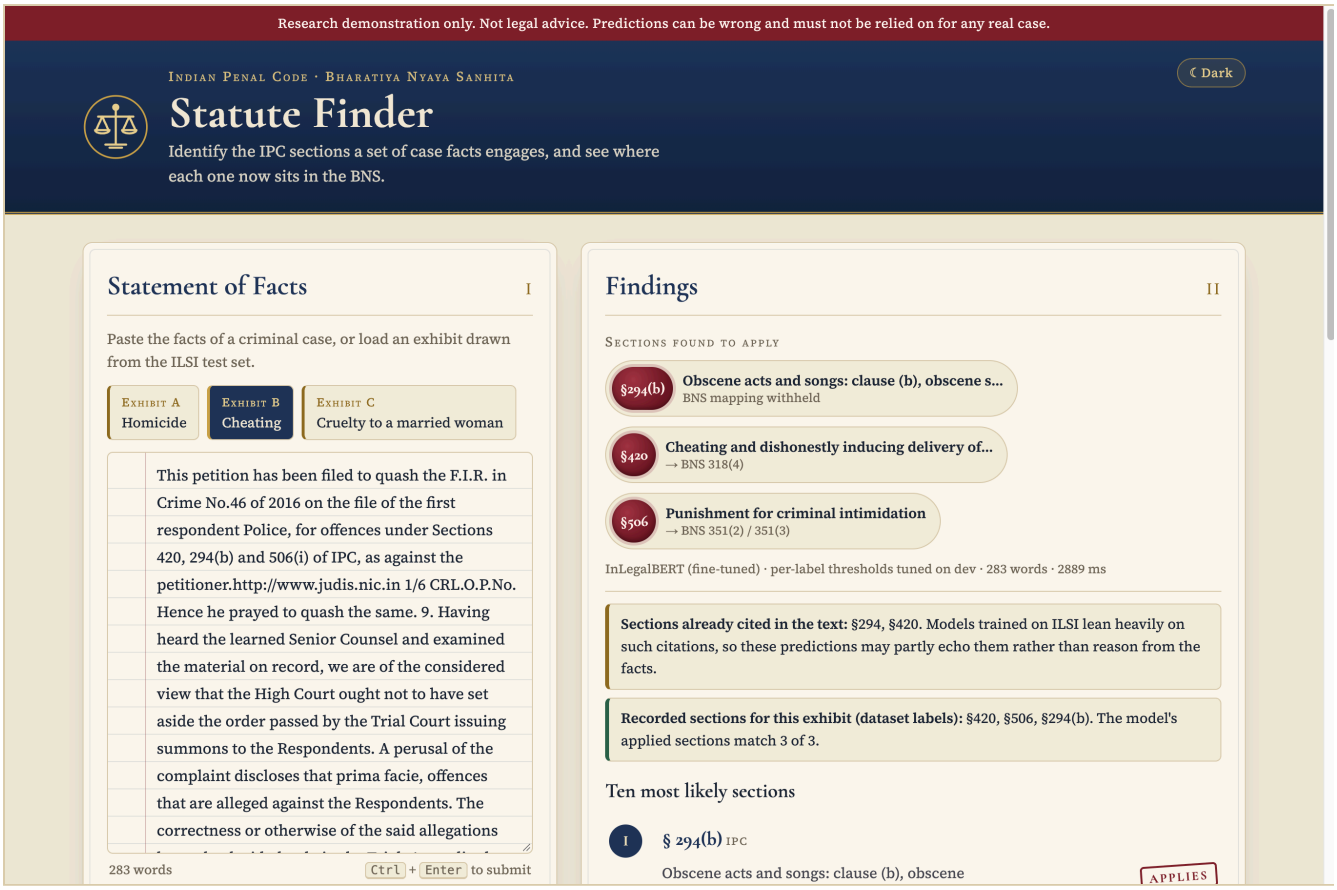


Figure 15. *The same view in the light theme.*

9. Limitations and ethics

Limitations

- **Benchmark leakage.** ILSI labels can largely be recovered from citations in the text, so headline scores overstate how well any model infers law from facts. The "never cited" subset (19% of test) is the more honest measure.
- **Label noise.** Non-offence sections, repealed sections and apparently mis-linked CrPC citations appear as labels, so some "errors" are label errors.
- **Reduced search.** One training seed per BERT configuration, a reduced regularisation grid, no SVM or character n-gram model, and 512-token truncation. Small differences such as head vs head+tail (−0.9 points) are within what seed variance could explain; this was not measured.
- **Possible pre-training overlap (unverified).** InLegalBERT was pre-trained on Indian Supreme Court and High Court judgments from 1950–2019, which may include ILSI test judgments.
- **Mapping source.** The BNS mapping rests on a secondary source; "verified" means checked against that table, not against an official government publication.

Ethics

- **Not legal advice.** The demo says so prominently, shows the mapping status, and warns when predictions may only echo citations.
- **Automation bias.** Statute suggestions can anchor investigators, lawyers or judges. A model trained on past charging and judicial practice reproduces its patterns, including any over-charging or bias

across communities. It should support human review, never replace it.
- **Privacy.** Judgments are public but name parties, victims and witnesses. The data should be used for research only and not re-published in identifying form.
- **The law has changed.** After 1 July 2024, an IPC-only model is wrong for new offences by construction (Section 7).

10. Conclusion and future work

On ILSI, a well-tuned TF-IDF + logistic regression model beats fine-tuned BERT models by a large, significant margin. The analysis shows why: most judgments cite their own answer, whole-document lexical models read those citations, and 512-token BERT models see only some of them and do not reliably copy the ones they see. Where no citation exists, the problem is genuinely hard for all models. Legal-domain pre-training (InLegalBERT) helps modestly, especially for rare sections. The move to the BNS makes citation-based models fragile, because section numbers change meaning and new offences have no training data.

Future work

1. **Mask section numbers** and other Acts' citations in training and test text, so models are evaluated on facts → offence. This is the most important follow-up.
2. **Longer context:** hierarchical or long-document legal encoders, or chunk-and-aggregate BERT, to lift the 512-token limit.
3. **Robustness:** multiple seeds with confidence intervals over training runs; the full regularisation grid and SVM baselines.
4. **Label cleaning:** remove non-offence and repealed sections, and filter citations from other Acts.
5. **Statute-aware methods:** retrieval and graph methods that use statute text (as LeSICiN does), and the BNS zero-shot experiment once BNS texts are available.
6. **A dual IPC/BNS model,** fine-tuned on BNS-era judgments as they appear.

11. References

1. S. Paul, P. Goyal, S. Ghosh. *LeSICiN: A Heterogeneous Graph-based Approach for Automatic Legal Statute Identification from Indian Legal Documents*. Proceedings of AAAI 2022. Dataset: Zenodo, doi:10.5281/zenodo.6053791.
2. S. Paul, A. Mandal, P. Goyal, S. Ghosh. *Pre-trained Language Models for the Legal Domain: A Case Study on Indian Law*. Proceedings of ICAIL 2023; arXiv:2209.06049. Model: huggingface.co/law-ai/InLegalBERT.
3. J. Devlin, M.-W. Chang, K. Lee, K. Toutanova. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL 2019.
4. *The Bharatiya Nyaya Sanhita, 2023* (Act No. 45 of 2023), in force from 1 July 2024. India Code: indiacode.nic.in/handle/123456789/20062.
5. A.K. Yadav (Director, CAPT Bhopal). *Correspondence Table and Comparison Summary of the Bharatiya Nyaya Sanhita, 2023 to the Indian Penal Code, 1860*. keralaprisons.gov.in/userfiles/act-and-rules/comparison_summary_BNS_to_IPC.pdf (secondary source).

Appendix A: Hyperparameters and compute

Component	Setting
Data split	Official ILSI train / dev / test (42,835 / 10,200 / 13,039)
Keyword baseline	Statute-text TF-IDF, English stop words, alphabetic tokens; top-k = 5 (dev-selected from 5/10/15/25)
TF-IDF features	word 1–2-grams, max 100,000 features, sublinear tf, min_df 3, max_df 0.9, float32
Logistic regression	one-vs-rest, liblinear, class_weight = balanced, C = 1 (dev-selected from 0.5/1), 4 threads
BERT encoders	bert-base-uncased; law-ai/InLegalBERT
Sequence length	512 tokens; head (510) or head+tail (128 + 382)
Optimisation	AdamW, lr 2e-5, linear schedule, 10% warm-up, weight decay 0.01, 4 epochs, fp16
Batch	16 per step × 2 gradient accumulation = 32
Loss	BCE with logits, pos_weight = neg/pos clipped to [1, 30]
Model selection	best epoch by dev micro-F1 at 0.5
Thresholds	fixed 0.5; global grid 0.05–0.95 step 0.01; per-label (≥ 5 dev positives)
Seeds	42 everywhere
Compute	Baselines: Apple M1 (8 GB RAM), TF-IDF LR ≈ 30 min. BERT: Colab NVIDIA A100 40 GB, ≈ 13 min per run
Software	Python 3.11, PyTorch 2.14 (Mac) / 2.11 (Colab), transformers 5.20.2, datasets 5.1.0, scikit-learn 1.9.1

Table A1. BERT training log per epoch.

Run	Epoch	Train loss	Dev loss	Dev micro-F1 @0.5	Dev macro-F1 @0.5
InLegalBERT (head)	1	0.847	0.676	0.327	0.254
InLegalBERT (head)	2	0.592	0.598	0.379	0.306
InLegalBERT (head)	3	0.519	0.573	0.399	0.326
InLegalBERT (head)	4	0.481	0.564	0.396	0.324
BERT-base (head)	1	0.856	0.705	0.315	0.242
BERT-base (head)	2	0.612	0.616	0.349	0.275
BERT-base (head)	3	0.536	0.588	0.369	0.294
BERT-base (head)	4	0.496	0.583	0.371	0.297
InLegalBERT (head+tail)	1	0.853	0.690	0.318	0.246
InLegalBERT (head+tail)	2	0.602	0.611	0.370	0.300
InLegalBERT (head+tail)	3	0.528	0.586	0.384	0.312
InLegalBERT (head+tail)	4	0.490	0.580	0.389	0.316

Appendix B: Label set with BNS mapping

All 100 ILSI labels with their training frequency, frequency bucket and BNS counterpart. *verified = yes* means confirmed by an explicit row of the source table (or ILSI's statute text for repealed sections). Empty BNS cells are mappings withheld because they could not be verified. Notes and sources for every row are in `tables/ipc_to_bns_mapping.csv`.

ipc_section	ipc_title	train_docs	bucket	bns_sections	mapping_type	verified
2	Punishment of offences committed within India	343	tail	1(3)	one_to_one	yes
3	Punishment of offences committed beyond, but which by law may be tried within, India	1038	mid	1(4)	one_to_one	yes
4	Extension of Code to extra-territorial offences	367	tail	1(5)	one_to_one	yes
5	Certain laws not to be affected by this Act	454	tail	1(6)	one_to_one	yes
13	[Repealed] Definition of 'Queen'	341	tail		repealed	yes
34	Acts done by several persons in furtherance of common intention	11751	head	3(5)	one_to_one	yes
107	Abetment of a thing	699	mid	45	one_to_one	yes
109	Punishment of abetment if the act abetted is committed in consequence and where no express provision is made for its punishment	1040	mid	49	one_to_one	yes
114	Abettor present when offence is committed	350	tail	54	one_to_one	yes
120	Concealing design to commit offence punishable with imprisonment	823	mid	60	one_to_one	yes
120B	Punishment of criminal conspiracy	4141	head	61(2)	one_to_one	yes
143	Punishment (for being member of unlawful assembly)	617	mid		unclear	no
147	Punishment for rioting	4330	head	191(2)	one_to_one	yes
148	Rioting, armed with deadly weapon	4142	head	191(3)	one_to_one	yes
149	Every member of unlawful assembly guilty of offence committed in prosecution of common object	3822	head	190	one_to_one	yes
155	Liability of person for whose benefit riot is committed	338	tail	193(2)	one_to_one	yes
156	Liability of agent of owner or occupier for whose benefit riot is committed	665	mid	193(3)	one_to_one	yes
161	[Repealed] Public servant taking gratification	834	mid		repealed	yes
164	[Repealed] Punishment for abetting offence defined in section 162 or 163 by public servant	394	tail		repealed	yes
173	Preventing service of summons or other proceeding, or preventing publication thereof	524	tail	207	one_to_one	yes
174A	Non-appearance in response to a proclamation under section 82 of Act 2 of 1974	488	tail	209	one_to_one	yes
186	Obstructing public servant in discharge of public functions	684	mid	221	one_to_one	yes
188	Disobedience to order duly promulgated by public servant	528	tail	223	one_to_one	yes
190	Threat of injury to induce person to refrain from applying for protection to public servant	305	tail	225	one_to_one	yes
193	Punishment for false evidence	470	tail	229	one_to_one	yes
200	Using as true such declaration knowing it to be false	287	tail	237	one_to_one	yes
201	Causing disappearance of evidence of offence, or giving false information to screen offender	1911	head	238	one_to_one	yes
228	Intentional insult or interruption to public servant sitting in judicial proceeding	330	tail	267	one_to_one	yes
229A	Failure by person released on bail or bond to appear in Court	605	mid	269	one_to_one	yes

ipc_section	ipc_title	train_docs	bucket	bns_sections	mapping_type	verified
279	Rash driving or riding on a public way	498	tail	281	one_to_one	yes
294	Obscene acts and songs	3016	head	296	one_to_one	yes
299	Culpable homicide	385	tail	100	one_to_one	yes
300	Murder	1483	head	101	one_to_one	yes
302	Punishment for murder	10391	head	103(1)	one_to_one	yes
304	Punishment for culpable homicide not amounting to murder	3157	head	105	one_to_one	yes
304A	Causing death by negligence	569	mid	106(1)	one_to_one	yes
304B	Dowry death	1271	mid	80	one_to_one	yes
306	Abetment of suicide	1714	head	108	one_to_one	yes
307	Attempt to murder	7124	head	109	one_to_one	yes
308	Attempt to commit culpable homicide	544	mid	110	one_to_one	yes
313	Causing miscarriage without woman's consent	821	mid	89	one_to_one	yes
320	Grievous hurt	566	mid	116	modified	yes
323	Punishment for voluntarily causing hurt	7742	head	115(2)	one_to_one	yes
324	Voluntarily causing hurt by dangerous weapons or means	3662	head	118(1)	one_to_one	yes
325	Punishment for voluntarily causing grievous hurt	2619	head	117(2)	one_to_one	yes
326	Voluntarily causing grievous hurt by dangerous weapons or means	2513	head	118(2)	one_to_one	yes
332	Voluntarily causing hurt to deter public servant from his duty	881	mid	121(1)	one_to_one	yes
336	Act endangering life or personal safety of others	585	mid	125	one_to_one	yes
337	Causing hurt by act endangering life or personal safety of others	380	tail	125(a)	one_to_one	yes
338	Causing grievous hurt by act endangering life or personal safety of others	319	tail	125(b)	one_to_one	yes
341	Punishment for wrongful restraint	3186	head	126(2)	one_to_one	yes
342	Punishment for wrongful confinement	1086	mid	127(2)	one_to_one	yes
353	Assault or criminal force to deter public servant from discharge of his duty	1279	mid	132	one_to_one	yes
354	Assault or criminal force to woman with intent to outrage her modesty	2210	head	74	one_to_one	yes
363	Punishment for kidnapping	1787	head	137(2)	modified	yes
364	Kidnapping or abducting in order to murder	558	mid	140(1)	one_to_one	yes
365	Kidnapping or abducting with intent secretly and wrongfully to confine person	435	tail	140(3)	one_to_one	yes
366	Kidnapping, abducting or inducing woman to compel her marriage, etc.	1433	head	87	one_to_one	yes
366A	Procuration of minor girl	310	tail	96	modified	yes
375	Rape	358	tail	63	modified	yes
376	Punishment for rape	3728	head	64	one_to_one	yes
379	Punishment for theft	1662	head	303(2)	one_to_one	yes
380	Theft in dwelling house, etc.	635	mid	305	modified	yes
384	Punishment for extortion	353	tail	308(2)	one_to_one	yes
389	Putting person in fear of accusation of offence, in order to commit extortion	310	tail	308(7)	one_to_one	yes
392	Punishment for robbery	1093	mid	309(4)	one_to_one	yes
394	Voluntarily causing hurt in committing robbery	630	mid	309(6)	one_to_one	yes
395	Punishment for dacoity	644	mid	310(2)	one_to_one	yes
397	Robbery or dacoity, with attempt to cause death or grievous hurt	1362	head	311	one_to_one	yes

ipc_section	ipc_title	train_docs	bucket	bns_sections	mapping_type	verified
406	Punishment for criminal breach of trust	2305	head	316(2)	one_to_one	yes
409	Criminal breach of trust by public servant, or by banker, merchant or agent	1035	mid	316(5)	one_to_one	yes
411	Dishonestly receiving stolen property	592	mid	317(2)	one_to_one	yes
415	Cheating	319	tail	318(1)	one_to_one	yes
417	Punishment for cheating	486	tail	318(2)	one_to_one	yes
419	Punishment for cheating by personation	533	tail	319(2)	one_to_one	yes
420	Cheating and dishonestly inducing delivery of property	4057	head	318(4)	one_to_one	yes
427	Mischief causing damage to the amount of fifty rupees	1741	head		unclear	no
436	Mischief by fire or explosive substance with intent to destroy house, etc.	437	tail	326(g)	one_to_one	yes
437	Mischief with intent to destroy or make unsafe a decked vessel or one of twenty tons burden	577	mid	327(1)	modified	yes
438	Punishment for the mischief described in section 437 committed by fire or explosive substance	1322	mid	327(2)	one_to_one	yes
447	Punishment for criminal trespass	604	mid	329(3)	one_to_one	yes
448	Punishment for house-trespass	982	mid	329(4)	one_to_one	yes
450	House-trespass in order to commit offence punishable with imprisonment for life	272	tail	332(b)	one_to_one	yes
452	House-trespass after preparation for hurt, assault or wrongful restraint	1554	head	333	one_to_one	yes
457	Lurking house-trespass or house-breaking by night in order to commit offence punishable with imprisonment	365	tail	331(4)	one_to_one	yes
465	Punishment for forgery	609	mid	336(2)	one_to_one	yes
467	Forgery of valuable security, will, etc.	1667	head	338	one_to_one	yes
468	Forgery for purpose of cheating	2119	head	336(3)	one_to_one	yes
471	Using as genuine a forged document or electronic record	2073	head	340(2)	one_to_one	yes
482	Punishment for using a false property mark	1329	head	345(3)	one_to_one	yes
494	Marrying again during lifetime of husband or wife	310	tail	82(1)	one_to_one	yes
498	Enticing or taking away or detaining with criminal intent a married woman	564	mid	84	modified	yes
498A	Husband or relative of husband of a woman subjecting her to cruelty	3840	head	85	one_to_one	yes
500	Punishment for defamation	377	tail	356(2)	one_to_one	yes
504	Intentional insult with intent to provoke breach of the peace	1982	head	352	modified	yes
506	Punishment for criminal intimidation	7899	head	351(2); 351(3)	one_to_many	yes
509	Word, gesture or act intended to insult the modesty of a woman	491	tail	79	one_to_one	yes
511	Punishment for attempting to commit offences punishable with imprisonment for life or other imprisonment	674	mid	62	one_to_one	yes
294(b)	Obscene acts and songs: clause (b), obscene songs/words in or near a public place	1139	mid		unclear	no
376(2)	Punishment for rape: sub-section (2), aggravated rape	473	tail		unclear	no

Appendix C: BNS provisions with no IPC equivalent

These provisions are marked as new in the source table; none has training data in ILSI.

bns_section	title	status
48	Abetment outside India for offence in India	new provision

bns_section	title	status
69	Sexual intercourse by employing deceitful means etc. (incl. false promise of marriage)	new offence
95	Hiring, employing or engaging a child to commit an offence	new offence
103(2)	Murder by a group of five or more on grounds of race, caste, community, sex, place of birth, language, personal belief ('mob lynching')	new sub-clause
106(2)	Causing death by rash and negligent driving and escaping without reporting ('hit-and-run')	new; table notes it was kept on hold, not in force from 1 July 2024
111	Organised crime	new offence
112	Petty organised crime	new offence
113	Terrorist act	new offence
117(3)	Grievous hurt causing permanent disability or persistent vegetative state	new sub-clause
117(4)	Grievous hurt by a group of five or more on grounds of race, caste, etc.	new sub-clause
152	Act endangering sovereignty, unity and integrity of India	described as newly added in the table (often discussed as replacing IPC 124A sedition; not in the ILSI label set)
197(1)(d)	Making or publishing false or misleading information jeopardising the sovereignty, unity, integrity or security of India	new clause
226	Attempt to commit suicide to compel or restrain exercise of lawful power	new offence
304	Snatching	new offence
324(3)	Mischief causing loss or damage to any property including Government or local authority property	new sub-section

Appendix D: Reproducibility and file index

The project repository contains all code. From a fresh checkout with Python 3.11:

```
make setup        # virtual environment, pinned packages, Jupyter kernel
make data         # download ILSI (MD5-checked), inspect, build parquet files
make eda          # exploratory notebook
make baselines    # keyword, citation diagnostic, TF-IDF + LR
make colab-pack   # bundle for Colab; then run colab/train_bert.ipynb (3 runs)
make compare      # validate all outputs, master table, analysis notebook
make bns          # IPC -> BNS mapping, coverage, zero-shot retrieval
make report       # regenerate report figures, tables and REPORT.pdf
make app          # web demo at http://127.0.0.1:7860
make test         # unit tests
```

Contents of this report folder

- `REPORT.md` / `REPORT.pdf`: this report.
- `bns_discussion.md`: the full IPC → BNS discussion.
- `figures/`: all figures (EDA, results, analysis, BNS, application screenshots).
- `tables/`: every table as CSV, plus Markdown versions:
  - results (`results_main`, `results_all_strategies`), `bucket_f1`, `per_label_f1`, `citation_position`, `bootstrap`, `training_log`;
  - confusion and error-case tables;
  - the full IPC → BNS mapping, the new BNS offences, and the label set.